Stability thresholds of N-vortex rings on the hyperbolic plane (tanh model): rational Fourier blocks, palindromic threshold polynomials, and a certified table for $N \le 16$

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Abstract

We consider N identical point vortices on the hyperbolic plane $\mathbb{H}^2$ (curvature -1) with the pair interaction $-\ln\tanh(d/2)$, which is the Green's function of the Laplace–Beltrami operator that vanishes at infinity. Secondary sources attribute this Hamiltonian to Kimura (1999). Montaldi and Tokieda called the stability of the regular N-ring in this model "no longer tractable analytically". We show that in log-polar coordinates the Hessian of the augmented Hamiltonian at the ring splits into 2×2 Hermitian Fourier blocks whose entries are *rational functions* of $\xi = \tanh^2(r/2)$, where r is the hyperbolic ring radius. Unlike the "sinh" model, the blocks have a radial–tangential coupling. The blocks are invariant under $\xi \mapsto 1/\xi$. Consequently the linear-stability threshold $\xi_*(N)$ is an explicit algebraic number, the root of an exact rational equation. For $N = 8, \dots, 16$ we certify $\xi_*(N)$ by exact rational root isolation combined with Arb ball arithmetic. We also certify the critical mode $\lfloor N/2 \rfloor$, the positivity of the tangential entries, and the minimal polynomials of $\xi_*(N)$. These are irreducible and palindromic, and $\xi_*(8)$ is an algebraic unit of degree 10. Rings with $N \le 7$ are stable at every radius. For odd N, the energy–momentum (Hessian-definiteness) threshold lies strictly above the spectral threshold. The gap between them is a Krein window. Numerically, $\cosh r_* \sim \kappa_\infty N$ with $\kappa_\infty \approx 0.1343122$, whereas $\cosh r_* \sim \sqrt{N}/2$ in the sinh model.

1 Scope and status labels

The companion note `sinh/note.tex` treats the sinh model, where everything is known [1]. We use these labels:

PROVED (all N)	Fourier-block theorem (Thm. 2); linearisation and stability criteria (Prop. 3); reciprocity $a_m(1/\xi) = a_m(\xi)$ (Prop. 5); threshold equation and even-N form.
CERTIFIED for $N \le 16$	$b_m > 0$ on $(0,1)$; critical mode $\lfloor N/2 \rfloor$; the value of $\xi_*(N)$ and the sign change of a_{crit} at ξ_*; irreducible *palindromic* minimal polynomials and their degree pattern; $\xi_*(8)$ a unit; $N \le 7$ stable for all r; odd-N energy–momentum threshold strictly above ξ_*.
Lean-verified	In `lean/Havelock/TanhAlgebra.lean`, with no unproved steps and only the standard axioms. Taking the block formulas of Thm. 2 as definitions, Lean proves the following. (i) Reciprocity and the even-N threshold equation. (ii) For every $2 \le N \le 16$, the statements of Thm. 6 on positivity, sign change and critical mode, with ξ_* located to within 10^{-20} in a rational interval. (iii) $\xi_*(8)$ is an algebraic integer and a unit. (iv) The odd-N Krein window is non-empty for $N = 9, 11, 13, 15$. Lean does *not* verify irreducibility (hence the degrees), the derivation of the blocks, or the dynamics.
CONJECTURE	the degree pattern (4) and irreducibility/palindromy for all N; $b_m > 0$ and critical mode $\lfloor N/2 \rfloor$ for all N.
NUMERICAL	large-N constant κ_∞ and the fits in §8; critical mode up to $N = 800$.

Nonlinear (Lyapunov) stability is claimed only in the form "positive-definite Hessian on the non-rotational modes". The passage from this to Lyapunov stability holds *modulo* $SO(2,1)$; see Remark 4. We have not checked the hypotheses of an energy–momentum theorem for this model against a source.

2 The model

2.1 Hamiltonian

We use the Poincaré disk $\mathbb{D} = \{|z| < 1\}$ with the metric $4|dz|^2/(1-|z|^2)^2$ and the area (symplectic) form $\omega = \sum_k \lambda(z_k)\, dx_k \wedge dy_k$, $\lambda(z) = 4/(1-|z|^2)^2$. The geodesic distance satisfies $\tanh(d(z,w)/2) = |z-w|/|1-z\bar{w}|$. For N unit vortices, take

$$H = -\sum_{j<k} \ln\tanh\frac{d_{jk}}{2} = \sum_{j<k}\Big(-\ln|z_j - z_k| + \ln|1 - z_j\bar{z}_k|\Big), \qquad J = \sum_k \frac{1+|z_k|^2}{1-|z_k|^2} = \sum_k \cosh r_k. \quad (1)$$

Here J generates rotations about 0, and Hamilton's equations are $\dot{x}_k = \lambda^{-1}\partial_{y_k} H$, $\dot{y}_k = -\lambda^{-1}\partial_{x_k} H$, as in `note.tex`. The function $f(d) = \ln\tanh(d/2)$ satisfies $f'' + \coth d\, f' = 0$ for $d > 0$ (since $f' = 1/\sinh d$). So $G(d) = -\frac{1}{2\pi}\ln\tanh(d/2)$ is the Green's function of $\Delta_{\mathbb{H}^2}$ that vanishes as $d \to \infty$.

Attribution (secondary only). Montaldi–Tokieda [1, §4.2] state that the image Green's function $\log|z-w|^2/|1+\lambda z\bar{w}|^2$ "on the hyperbolic plane $\lambda < 0 \ldots$ reduces to the Hamiltonian adopted by Kimura". Montaldi–Nava-Gaxiola [2, eq. (12)] write Kimura's Hamiltonian as $-\frac{1}{4\pi}\sum \Gamma_i\Gamma_j \ln\frac{\langle X_i, X_j\rangle + 1}{\langle X_i, X_j\rangle - 1}$ with $\langle X, Y\rangle = -\cosh d$, which equals $-\frac{1}{4\pi}\sum \ln\tanh^2(d/2)$. We have *not* read Kimura [3]. Hence

(1) is "the model attributed to Kimura", and we quote nothing from Kimura directly. Montaldi–Tokieda add that the stability calculations for this choice "get so complicated" that they are "no longer tractable analytically" [1, §4.2].

No Robin term (intrinsic remark). $\mathbb{H}^2$ is homogeneous, so the regular part of G at the diagonal, measured in the intrinsic metric, is the constant $\lim_{d\to 0}(G(d) + \frac{1}{2\pi}\ln d) = \frac{1}{2\pi}\ln 2$. The Robin (self-interaction) function is therefore constant, and (1) has no one-body self-term. This contrasts with the flat disk with a hard wall (§9), where the Robin function $\frac{1}{2\pi}\ln(1-|z|^2)$ is not constant and, combined with the Euclidean area form, changes the dynamics. In conformal coordinates, a one-body term $-\frac{1}{2}\sum \ln(1-|z_k|^2)$ does appear in (2) below. That term comes from the $j \neq k$ image sums, not from a self-interaction.

Relation to the sinh model. Since $\sinh(d/2) = \tanh(d/2)\cosh(d/2)$, $H_{\sinh} = -\sum \ln\sinh(d_{jk}/2) = H - \sum_{j<k}\ln\cosh(d_{jk}/2)$. A direct computation gives $\Delta_{\mathbb{H}^2}\ln\cosh(d/2) = \frac{1}{2}$. So the sinh model is the tanh model plus a uniform compensating background vorticity attached to each vortex. This is the continuation to $\mathbb{H}^2$ of the sphere model, where Gauss's law forces such a term. The sinh model is the $\lambda < 0$ member of Montaldi–Tokieda's family [1, (1.12)].

2.2 Power-sum decomposition

Let $P_p = \sum_j z_j^p$. Using $\ln|1-u| = -\operatorname{Re}\sum_{p\geq 1} u^p/p$ and $\sum_{j\neq k}(z_j\bar{z}_k)^p = |P_p|^2 - \sum_j |z_j|^{2p}$, we get

$$H = -\sum_{j<k}\ln|z_j - z_k| \;-\; \frac{1}{2}\sum_j \ln(1-|z_j|^2) \;-\; \frac{1}{2}\sum_{p\geq 1}\frac{|P_p|^2}{p}. \tag{2}$$

The series converges absolutely on $\mathbb{D}^N$. The last term $\Phi := -\frac{1}{2}\sum_p |P_p|^2/p$ is the source of the apparent intractability. On a ring it is diagonal in the Fourier modes.

3 Fourier blocks

3.1 Coordinates and the relative equilibrium

Let $w = e^{2\pi i/N}$ and write $z_j = \sqrt{\xi}\, w^j e^{s_j + i\theta_j}$, so the ring of Euclidean radius $\sqrt{\xi}$ is $s = \theta = 0$. Then $\omega = \sum_j \kappa(s_j)\, ds_j \wedge d\theta_j$ with $\kappa(s) = 4\xi e^{2s}/(1-\xi e^{2s})^2$, and $\kappa_0 := \kappa(0) = 4\xi/(1-\xi)^2$. Set

$$c := \cosh r = \frac{1+\xi}{1-\xi}, \qquad X := \xi^N, \qquad W_k := \sum_{\substack{p\geq 1 \\ p\equiv k \ (N)}} p\,\xi^p = \frac{k\xi^k}{1-X} + \frac{N\xi^{k+N}}{(1-X)^2} \quad (1\leq k\leq N).$$

In particular $W_N = NX/(1-X)^2$. Let $F = H - \Omega J$.

Lemma 1. *The ring is a relative equilibrium for every $\xi \in (0,1)$, with $\Omega\kappa_0 = -\dfrac{N-1}{2} + \dfrac{\xi}{1-\xi} - \dfrac{NX}{1-X}$. At the ring, $dF = 0$ in all $2N$ variables.*

Proof. All θ-derivatives vanish by the dihedral symmetry. For s_j: the pair term contributes $-\frac{N-1}{2}$ (its linear part is $-\frac{1}{2}\sum_{j<k}(s_j+s_k)$; see `note.tex`, §2). The one-body part is $V(s) = -\frac{1}{2}\ln(1-u) - \Omega\frac{1+u}{1-u}$ with $u = \xi e^{2s}$, and $V'(0) = \frac{\xi}{1-\xi} - \Omega\kappa_0$. For Φ, write $|P_p|^2 = \xi^p\sum_{j,k} w^{(j-k)p} e^{p\epsilon_j + p\bar{\epsilon}_k}$ with $\epsilon_j = s_j + i\theta_j$. Then $\partial_{s_j}|P_p|^2\big|_0 = 2p\,\xi^p \operatorname{Re}\sum_k w^{(j-k)p} = 2pN\xi^p[N \mid p]$, so $\partial_{s_j}\Phi|_0 = -N\sum_{q\geq 1} X^q = -NX/(1-X)$. Setting the sum to zero gives Ω. □

3.2 The block theorem

Theorem 2 (Fourier blocks; PROVED)**.** *In the coordinates (s,θ), the Hessian of F at the ring is block-circulant. On the Fourier mode $(s_j,\theta_j)=(S,T)\,w^{mj}$, $1\le m\le N-1$, it acts as the Hermitian matrix*

$$M_m=\begin{pmatrix} a_m & ig_m\\ -ig_m & b_m\end{pmatrix},$$

where

$$\begin{aligned} a_m &= (N-1)c-\tfrac12(c-1)^2-\tfrac{m(N-m)}{2}+\frac{2NcX}{1-X}-\frac{N^2X}{(1-X)^2}\\ &\quad-\frac{N}{2}\,\frac{m\xi^m+(N-m)\xi^{N-m}+(N-m)\xi^{N+m}+m\xi^{2N-m}}{(1-X)^2},\\ b_m &= \tfrac{m(N-m)}{2}+NW_N-\tfrac{N}{2}\,(W_m+W_{N-m}),\qquad g_m=\tfrac{N}{2}\,(W_m-W_{N-m}).\end{aligned}$$

On mode 0, $b_0=g_0=0$. All entries are rational functions of ξ with rational coefficients. Moreover $a_{N-m}=a_m$, $b_{N-m}=b_m$, $g_{N-m}=-g_m$, and $g_{N/2}=0$ for even N.

For comparison, the sinh model has $a_m=(N-1)(1+c^2)/2-m(N-m)/2$, $b_m=m(N-m)/2$ and $g_m=0$ (`note.tex`, Prop. 2).

Proof. Since $dF=0$ at the ring (Lemma 1), the Hessian is well defined. We add the three contributions in (2) and the term $-\Omega J$.

Pair term. As in `note.tex` (Lemma 1 and the expansion before it), $-\sum_{j<k}\ln|z_j-z_k|$ has Hessian $-\frac{m(N-m)}{2}$ in ss and $+\frac{m(N-m)}{2}$ in $\theta\theta$ on mode m, with no $s\theta$ coupling.

One-body terms. $V(s)$ from the proof of Lemma 1 does not depend on θ. Using $\frac{d}{ds}=2u\frac{d}{du}$,

$$V''(0)=\frac{2\xi}{(1-\xi)^2}-\frac{8\Omega\xi(1+\xi)}{(1-\xi)^3}=\frac{2\xi}{(1-\xi)^2}-2c\,\Omega\kappa_0.$$

Substituting $\Omega\kappa_0$ and using $c-1=2\xi/(1-\xi)$, so that $\frac{2\xi}{(1-\xi)^2}-\frac{2c\xi}{1-\xi}=-\frac12(c-1)^2$, gives $V''(0)=(N-1)c-\frac12(c-1)^2+\frac{2NcX}{1-X}$. This term is the same on every mode and lies in the ss entry.

Power sums. The second-order part of $e^{p\epsilon_j+p\bar\epsilon_k}$ is $\frac{p^2}{2}(\epsilon_j^2+2\epsilon_j\bar\epsilon_k+\bar\epsilon_k^2)$. Summing with the weight $w^{(j-k)p}$: (i) the squares give $p^2N[N\mid p]\sum_j\mathrm{Re}\,\epsilon_j^2=p^2N[N\mid p]\sum_j(s_j^2-\theta_j^2)$; (ii) the cross term gives $p^2\big|\sum_j w^{jp}\epsilon_j\big|^2$. Hence the quadratic part of Φ is

$$\Phi_2=-\frac{N}{2}W_N\sum_j(s_j^2-\theta_j^2)\;-\;\frac12\sum_{p\ge1}p\,\xi^p\Big|\sum_j w^{jp}(s_j+i\theta_j)\Big|^2.$$

The first piece contributes $-NW_N$ to ss and $+NW_N$ to $\theta\theta$. For the second, expand $|\sum_j w^{jp}\epsilon_j|^2$ as a real quadratic form. It equals $\sum_{j,k}\cos\phi_{jk}(s_js_k+\theta_j\theta_k)+2\sum_{j,k}\sin\phi_{jk}\,s_j\theta_k$ with $\phi_{jk}=2\pi(j-k)p/N$. Its Hessian has circulant blocks $2\cos\phi_{jk}$ (ss, $\theta\theta$) and $2\sin\phi_{jk}$ ($s\theta$). On the vector w^{mk} their symbols are $N([p\equiv m]+[p\equiv -m])$ and $-iN([p\equiv m]-[p\equiv -m])$, using $\sum_k w^{(j-k)p}w^{mk}=Nw^{mj}[p\equiv m]$. Multiplying by $-\frac12 p\xi^p$ and summing over p gives $-\frac{N}{2}(W_m+W_{N-m})$ on both diagonal entries and $+\frac{iN}{2}(W_m-W_{N-m})=ig_m$ in the $s\theta$ entry.

Adding the three pieces gives a_m, b_m and g_m. In particular, $a_m=V''(0)-\frac{m(N-m)}{2}-NW_N-\frac{N}{2}(W_m+W_{N-m})$, and expanding W_m+W_{N-m} over the common denominator $(1-X)^2$ gives the bracket in the statement. For $m=0$, W_0 should be read as W_N, which gives $b_0=NW_N-NW_N=0$. The symmetries follow from $m\leftrightarrow N-m$. □

Checks. The blocks equal the Fourier symbol of the exact Cartesian Hessian of F (sympy second derivatives, with Ω solved numerically) to $\le 2\cdot 10^{-48}$. The grid was $N \in \{3,5,7,8,9,10,12,13,16\}$ and $\xi \in \{0.02, 0.09, 0.3, 0.75\}$ (`check_jacobian.out`). An independent second reader built a Cartesian Hessian at 60–110 digits without seeing these formulas and matched the blocks to $2.5\cdot 10^{-59}$ (`review/`). The compact form of a_m agrees exactly with the block formula for $N = 3..16$ and all m (`identities.out`).

4 Linearisation and stability criteria

Proposition 3 (PROVED)**.** *On mode m the linearised flow is $\kappa_0(\dot S, \dot T)^\top = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) M_m (S,T)^\top$. Its eigenvalues are*

$$\kappa_0 \sigma = -i g_m \pm \sqrt{-a_m b_m}.$$

Hence the ring is spectrally stable iff $a_m b_m \ge 0$ for all $1 \le m \le N-1$. The Hessian of F is positive definite on the non-rotational modes $1 \le m \le N-1$ iff $a_m > 0$ and $a_m b_m - g_m^2 > 0$ for all such m.

Proof. $\det\left(\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) M_m - \lambda\right) = (\lambda + i g_m)^2 + a_m b_m$. A 2×2 Hermitian matrix is positive definite iff its $(1,1)$ entry and its determinant $a_m b_m - g_m^2$ are positive. Mode 0 (rotation, and change of ξ along the family) carries $b_0 = 0$ and is excluded. □

So the coupling g_m only shifts the frequencies; it does not affect spectral stability. It does affect definiteness, and this is the origin of the odd-N window in §7.

Remark 4 (energy–momentum, "modulo $SO(2,1)$"). The variation dJ annihilates every mode $m \neq 0$. Hence $\ker dJ \subset (\text{modes } 1..N-1) \oplus \mathbb{R}\,\partial_\theta^{(0)}$, and positive definiteness on modes $1..N-1$ implies definiteness of d^2F on $\ker dJ$ modulo the rotation orbit. The full symmetry group of (1) is the isometry group $PSL(2,\mathbb{R}) \cong SO(2,1)_0$. Its momentum map has three components, and the modes $m = \pm 1$ are tangent to the non-compact (boost) directions. We therefore state any Lyapunov conclusion only *modulo* $SO(2,1)$: up to motion along the group orbit. We do not claim stability modulo $SO(2)$ alone. We did not verify the hypotheses of a specific energy–momentum theorem (for example Montaldi–Tokieda [1, Thm. 2], which is stated for the sinh model) for the tanh model.

5 Reciprocity and the threshold equation

Proposition 5 (reciprocity; PROVED for all N, m)**.** *As rational functions, $a_m(1/\xi) = a_m(\xi)$.*

Proof. Under $\xi \mapsto 1/\xi$ we have $c \mapsto -c$ and $X \mapsto 1/X$. The one-body part changes as follows: $(N-1)c \mapsto -(N-1)c$; $-\frac{1}{2}(c-1)^2 \mapsto -\frac{1}{2}(c+1)^2$, a change of $-2c$; and $\frac{2NcX}{1-X} \mapsto \frac{-2Nc/X}{1-1/X} = \frac{2Nc}{1-X}$, a change of $+2Nc$. The total change is $-2(N-1)c - 2c + 2Nc = 0$. The term $N^2X/(1-X)^2$ is invariant. In the last fraction, the numerator is multiplied by ξ^{-2N} (its exponents $m, N-m, N+m, 2N-m$ are permuted by $e \mapsto 2N-e$) and so is $(1-X)^2$. The term $m(N-m)/2$ is constant. □

A symbolic check treating ξ, $X = \xi^N$ and $t = \xi^m$ as independent symbols confirms the identity (`review/palin_symbolic.py`). Clearing denominators, $A_m := 2(1-\xi)^2(1-X)^2 a_m \in \mathbb{Z}[\xi]$ satisfies $A_m(1/\xi) = \xi^{-2N-2} A_m(\xi)$, so A_m is self-reciprocal. We write $B_m := 2(1-X)^2 b_m$, $G_m := 2(1-X)^2 g_m$ and $D_m := A_m B_m - (1-\xi)^2 G_m^2 = 4(1-\xi)^2(1-X)^4(a_m b_m - g_m^2)$. On $(0,1)$ the signs of a_m, b_m and $a_m b_m - g_m^2$ are those of A_m, B_m and D_m.

Threshold equation. If $b_m > 0$ on $(0,1)$ for all m (certified for $N \le 16$, §6), instability occurs only through the radial entries. The spectral threshold $\xi_*(N)$ is then the largest root in $(0,1)$ of $\prod_m a_m$. For $N \le 16$ this is the unique root of $a_{\lfloor N/2 \rfloor}$ in $(0,1)$. For even N and $m = N/2$, the bracket and the N^2X term combine. With

$$y := \xi^{N/2} = \tanh^N(r/2),$$

the threshold equation $a_{N/2} = 0$ reads

$$(N-1)c - \frac{(c-1)^2}{2} + \frac{2Ncy^2}{1-y^2} = \frac{N^2}{8} + \frac{N^2 y}{2(1-y)^2}, \qquad c = \frac{1+\xi}{1-\xi}. \tag{3}$$

(Use $m\xi^m + (N-m)\xi^{N-m} = Ny$, $(N-m)\xi^{N+m} + m\xi^{2N-m} = Ny^3$, and $y(1+y^2)+2y^2 = y(1+y)^2$.) For odd N we use the general a_m with $m = (N-1)/2$. In the sinh model the right-hand side is the same Thomson term $N^2/8$ without the image correction, and the left-hand side is $(N-1)(1+c^2)/2$.

The natural invariant. By reciprocity, if the minimal polynomial of ξ_* is palindromic, then $C := \xi_* + \xi_*^{-1}$ has half the degree. Since $\xi = \tanh^2(r/2)$,

$$C = \xi_* + \xi_*^{-1} = 2 + \frac{4}{\sinh^2 r_*} = \frac{2(\cosh^2 r_* + 1)}{\cosh^2 r_* - 1}.$$

So $\cosh^2 r_*$ generates the same field as C. In the sinh model, $C = 2 + 4/(q-1)$ is rational (`note.tex`, §4).

6 Certified thresholds for $N \le 16$

Method (`certify_v2.py` → `certify_v2.out`). For each $N \le 16$ and $1 \le m \le \lfloor N/2 \rfloor$:

1. Form the exact integer polynomials A_m, B_m, D_m (sympy, from Thm. 2).
2. Isolate all real roots in $(0,1)$ of the square-free parts with exact rational real-root isolation, after removing the factors ξ and $\xi - 1$. This shows: B_m has no root and $B_m(\frac{1}{2}) > 0$; every A_m with $m \ne \lfloor N/2 \rfloor$ has all its roots strictly below the isolating interval of $A_{\lfloor N/2 \rfloor}$; and $A_{\lfloor N/2 \rfloor}$ has exactly one root in $(0,1)$.
3. Factor A_{crit} over $\mathbb{Q}$ and select the unique irreducible factor f with a root in $(0,1)$. Check its irreducibility and refine an exact rational isolating interval $[\alpha, \beta]$ for its root, of width $< 10^{-45}$.
4. Assert exactly that $A_{\mathrm{crit}}(\alpha) < 0 < A_{\mathrm{crit}}(\beta)$. With step 2 this proves $a_{\mathrm{crit}} < 0$ on $(0, \xi_*)$ and $a_m > 0$ for all m on $(\xi_*, 1)$.
5. Compute Arb balls for the roots of f (python-flint 0.9, 400 bits). Select the real root by exact tests only: the real roots returned by `arb_fmpz_poly_complex_roots` have imaginary part exactly 0. Assert that the whole ball $[\mathrm{mid} - \mathrm{rad}, \mathrm{mid} + \mathrm{rad}]$, converted exactly to rationals, lies inside $[\alpha, \beta]$.
6. For the energy–momentum threshold, take the largest root of $\prod_m D_m$ in $(0,1)$. Isolate it exactly, assert that it is separated from the roots of the other D_m, and compare it exactly with $[\alpha, \beta]$.

No floating-point comparison enters any pass/fail decision. The first version (`certify.py`) selected the Arb root by a float comparison of its midpoint. The second reader flagged this, and version 2 replaces it. All values and polynomials in v1 and v2 agree.

Theorem 6 (CERTIFIED, $2 \le N \le 16$)**.** *(a) For $N \le 7$, a_m, b_m and $a_m b_m - g_m^2$ are positive on all of $(0,1)$ for every m. The ring is spectrally stable and has positive-definite non-rotational Hessian at every radius.*

(b) For $8 \le N \le 16$, $b_m > 0$ on $(0,1)$ for every m. The ring is linearly unstable for $0 < \xi < \xi_*(N)$ and spectrally stable for $\xi_*(N) < \xi < 1$. The instability at threshold is in the mode $m = \lfloor N/2 \rfloor$ (and $N - m$), and every other a_m changes sign strictly below ξ_*.

(c) $\xi_*(N)$ is the root in $(0,1)$ of an irreducible palindromic integer polynomial of degree

$$\deg \xi_*(N) = \begin{cases} 2N-2, & N \text{ odd}, \\ \tfrac{3}{2}N - 2, & N \equiv 0 \pmod 4, \\ \tfrac{3}{2}N - 3, & N \equiv 2 \pmod 4, \end{cases} \tag{4}$$

with the values in Table 1.

N	ξ_*	$\sqrt{\xi_*}$	r_*	$\cosh r_*$	deg	lead
8	0.06735622300408122622684374…	0.2595307747	0.5312104560	1.1444414784	10	1
9	0.11287106323660274043969 40…	0.3359628897	0.6990695396	1.2544637167	16	4
10	0.16611424152623911808175 86…	0.4075711490	0.8653901256	1.3984100696	12	7
11	0.20460117688266929063553 58…	0.4523286160	0.9752480466	1.5144618547	20	10
12	0.24741777680406770917396 50…	0.4974110743	1.0917203545	1.6575169309	16	7
13	0.27986138864390668702999 65…	0.5290192706	1.1775646302	1.7772431147	24	18
14	0.31493896311550662339350 54…	0.5611942294	1.2691494599	1.9194478920	18	23
15	0.34245858351217025660102 26…	0.5851996100	1.3406681986	2.0416335000	28	28
16	0.37168181390604545663672 87…	0.6096571282	1.4167509599	2.1831006077	22	17

Table 1: Certified spectral thresholds of the tanh model. $\xi_* = \tanh^2(r_*/2)$ is the squared Euclidean radius in the Poincaré disk, r_* is the hyperbolic radius, "deg" is the degree of the minimal polynomial of ξ_* and "lead" is its leading coefficient. The Arb radii are $< 3 \cdot 10^{-153}$. The full minimal polynomials, those of $C = \xi_* + 1/\xi_*$, and the discriminants are in `certify_v2.out`.

Independent confirmation. The second reader recomputed $\xi_*(8), \xi_*(9), \xi_*(10), \xi_*(12)$ from full $2N$-dimensional Cartesian spectra and found agreement to ≥ 43 digits. A grid scan of the full Jacobian for $N = 3..7$ found $\max |\mathrm{Re}\,\sigma| \le 4 \cdot 10^{-19}$ (NUMERICAL).

Structure of the numerator. For even $N \in \{8, \dots, 16\}$, $A_{N/2}$ factors over $\mathbb{Q}$ as a constant times $(\xi - 1)^4(\xi^{N/2} + 1)$, times one further factor $(\xi + 1)$ when $N \equiv 2 \pmod 4$, times the minimal polynomial of ξ_*. So $(\xi + 1)^2$ divides $A_{N/2}$ exactly when $N \equiv 2 \pmod 4$, with one copy of $(\xi + 1)$ coming from $\xi^{N/2} + 1$. (An earlier draft counted this copy twice; the corrected form is checked in Lean, `A_crit_factor_N8–N16`.) These cyclotomic factors have roots on $|\xi| = 1$ and are spurious.

6.1 $N = 8$: an algebraic unit

Corollary 7 (CERTIFIED). *$\xi_*(8)$ is an algebraic unit of degree 10 with minimal polynomial*

$$x^{10} - 12x^9 - 37x^8 - 72x^7 - 84x^6 - 88x^5 - 84x^4 - 72x^3 - 37x^2 - 12x + 1.$$

$C = \xi_* + \xi_*^{-1} = 2 + 4/\sinh^2 r_*$ has minimal polynomial $C^5 - 12C^4 - 42C^3 - 24C^2 + 32C + 32$. For $9 \le N \le 16$ the leading coefficients are $4, 7, 10, 7, 18, 23, 28, 17$, so $\xi_*(N)$ is not an algebraic integer.

The polynomial is monic with constant term 1, so $\xi_*(8)$ is a unit. The second reader recovered the same polynomial independently by PSLQ from a 110-digit value and confirmed its irreducibility (`review/minpoly8.py`). Numerically, its only real roots are $\xi_*(8)$ and $1/\xi_*(8) \approx 14.846$. Four conjugates lie on $|x| = 1$ and four do not, so $\xi_*(8)$ is neither Pisot nor Salem. In the sinh model $\xi_*(8) = 8 - 3\sqrt{7}$ is also a unit, of degree 2. We do not know whether the coincidence has a reason.

Conjecture 8. *For every $N \ge 8$: $b_m > 0$ on $(0, 1)$; the critical mode is $\lfloor N/2 \rfloor$; the factor of $A_{\lfloor N/2 \rfloor}$ carrying ξ_* is irreducible and palindromic, so that $1/\xi_*$ is a Galois conjugate of ξ_*; and its degree is given by (4).*

Reciprocity alone shows that $1/\xi_*$ is a root of $A_{\lfloor N/2 \rfloor}$. It does not show that $1/\xi_*$ is a conjugate of ξ_*. That is why palindromy of the minimal polynomial is certified only for $N \le 16$. The critical mode and $b_m > 0$ near the threshold were checked numerically up to $N = 800$.

7 Odd N: the Krein window

For even N, $g_{N/2} = 0$, so $D_{N/2} = A_{N/2}B_{N/2}$ and the two thresholds coincide. This is certified: the same irreducible factor carries both. For odd N, $g_{\mathrm{crit}} \neq 0$. At ξ_* we have $a_{\mathrm{crit}} b_{\mathrm{crit}} - g_{\mathrm{crit}}^2 = -g_{\mathrm{crit}}^2 < 0$, and the Hessian becomes definite only at a larger value ξ_{EM}.

Proposition 9 (CERTIFIED, odd $9 \le N \le 15$). *$\xi_{\mathrm{EM}}(N) > \xi_*(N)$ strictly (disjoint exact rational intervals). The gaps are $3.220 \cdot 10^{-8}$ ($N = 9$), $1.244 \cdot 10^{-7}$ (11), $1.777 \cdot 10^{-7}$ (13) and $1.886 \cdot 10^{-7}$ (15). On $(\xi_*, \xi_{\mathrm{EM}})$ the ring is spectrally stable, with distinct purely imaginary eigenvalues, but the critical block M_{crit} has one positive and one negative eigenvalue.*

In the window the two frequencies of the critical block carry opposite Krein signatures. The second reader confirmed this on the coordinate-free Cartesian Hessian (count of negative eigenvalues, Krein signs of the eigenvectors; `review/krein_em.py`), so it is not a coordinate artefact. The energy–momentum method gives no conclusion there, and nonlinear stability in the window is **open**. In the sinh model $g_m \equiv 0$, and no such window exists.

8 Large N (NUMERICAL)

Set $c = \kappa N$. Then $\xi = (c-1)/(c+1)$ and $y = \xi^{N/2} \to u := e^{-1/\kappa}$. Dividing (3) by N^2 and formally letting $N \to \infty$ gives the limit equation

$$\kappa - \frac{\kappa^2}{2} + \frac{2\kappa u^2}{1 - u^2} - \frac{1}{8} - \frac{u}{2(1-u)^2} = 0, \qquad u = e^{-1/\kappa}. \tag{5}$$

Its root near 0.134 is

$$\kappa_\infty = 0.1343122142726281390560\ldots \qquad \text{(NUMERICAL; the limit equation is heuristic).}$$

Dropping the exponentially small u-terms gives $\kappa = 1 - \sqrt{3}/2 \approx 0.13397$. We have not proved that $\cosh r_*/N \to \kappa_\infty$: neither uniformity of the limit nor the odd-N case is proved. The data for $N \le 800$ (`asymptotics.out`) fit

$$\cosh r_* = \kappa_\infty N + \frac{0.541}{N} + O(N^{-2}), \qquad r_* = \ln(2\kappa_\infty N) + o(1), \qquad 1 - \xi_* \sim \frac{2}{\kappa_\infty N} \approx \frac{14.89}{N}.$$

The second reader re-solved (5) and checked $\cosh r_*/N$ up to $N = 1024$ (`review/kappa_check.py`).

9 Comparison with the sinh model and the flat disk

	sinh model [1]	tanh model (this note)
pair term	$-\ln\sinh(d/2)$	$-\ln\tanh(d/2)$
radial stiffness	$(N-1)(1+c^2)/2$	$(N-1)c - \frac{1}{2}(c-1)^2 + O(\xi^N)$
$s\theta$ coupling	none	$g_m = \frac{N}{2}(W_m - W_{N-m})$
threshold	$\cosh^2 r_* = q(N) \in \mathbb{Q}$	algebraic, degree (4)
$N \le 7$	stable for all r	stable for all r (cert.)
$\xi_*(8)$	$8 - 3\sqrt{7}$ (unit, deg 2)	unit, deg 10
large N	$\cosh r_* \sim \sqrt{N}/2$	$\cosh r_* \sim \kappa_\infty N$ (num.)
odd-N Krein window	none	width $\sim 10^{-7}$

For $8 \le N \le 16$ the tanh threshold is larger:

N	8	9	10	11	12	13	14	15	16
$r_*^{\sinh}$	0.5119	0.6585	0.7954	0.8814	0.9692	1.0317	1.0967	1.1462	1.1978
$r_*^{\tanh}$	0.5312	0.6991	0.8654	0.9752	1.0917	1.1776	1.2691	1.3407	1.4168

Both radii grow logarithmically, so $r_*^{\tanh}/r_*^{\sinh} \to 2$, but slowly. This is consistent with the data to $N = 800$ (NUMERICAL).

Interpretation. The models differ by the pair term $\ln\cosh(d/2)$, whose Laplacian is constant. In the sinh model this background gives a radial stiffness quadratic in $\cosh r$, which beats the Thomson destabilisation $N^2/8$ once $\cosh r \sim \sqrt{N}/2$. In the tanh model the interaction is screened by curvature: it decays like e^{-d} beyond one curvature radius. The stiffness is then only linear in $\cosh r$, up to exponentially small image terms, so stabilisation needs $\cosh r \sim 0.134N$.

Flat hard-wall disk (context only, NUMERICAL). This model has the Euclidean area form, image vortices, and the non-constant Robin self-image term. The reviewer scans (`checks/disk.py`, `checks/phys/bec.py`) find $N = 7, 8$ rings unstable at every radius tested, which README C5.3 reports as consistent with Havelock [5]. We have read neither Havelock nor Kurakin, and a Fourier-exact flat-disk analysis is not part of this note. The point is that the same image Green's function, with a different area form and a non-constant Robin function, gives qualitatively different thresholds. Neither hyperbolic model describes a flat-bottomed hard-wall condensate (README §3, item 1; retracted).

10 Limitations

- **Literature not read.** Kimura [3] and Hwang–Kim 2013 [4] (whose abstract says rings "are investigated in detail") could not be obtained (publisher 403). A human with journal access needs to check (i) which Hamiltonian Kimura uses (the tanh attribution here rests on [1, 2] and four further secondary sources) and (ii) whether Hwang–Kim 2013 already contains ring stability thresholds for this model. Until then, novelty is *unchecked*.
- **No human specialist review.** The derivation has been re-derived and checked by an independent second agent (verdict: confirmed with scope fixes, all applied here). No human expert in point-vortex dynamics has reviewed it.
- **Heuristic limit.** The equation (5) for κ_∞ comes from a formal limit. The constant and the fits in §8 are numerical.
- **Finite range.** Palindromy, irreducibility, the degree pattern, $b_m > 0$ and the critical mode are certified only for $N \le 16$ (Conjecture 8).
- **Nonlinear stability.** This is stated only via Hessian definiteness and modulo $SO(2,1)$ (Remark 4). It is open in the odd-N Krein window and at the threshold itself.

Files

`tanh/`: `derivation.txt` (full derivation), `blocks.py`, `exact.py`, `check_jacobian.py/.out`, `identities.py/.out` `certify_v2.py/.out` (certification), `asymptotics.py/.out`, and `review/` (second reader).

References

[1] J. Montaldi, T. Tokieda, Deformation of geometry and bifurcations of vortex rings, in *Recent Trends in Dynamical Systems*, Springer Proc. Math. Stat. 35 (2013) 335–370; arXiv:1210.5662.

[2] J. Montaldi, C. Nava-Gaxiola, Point vortices on the hyperbolic plane, J. Math. Phys. 55 (2014) 102702; arXiv:1403.2138.

[3] Y. Kimura, Vortex motion on surfaces with constant curvature, Proc. R. Soc. Lond. A 455 (1999) 245–259 (**not obtained**; cited only via [1, 2]).

[4] S. Hwang, S.-C. Kim, Relative equilibria of point vortices on the hyperbolic sphere, J. Math. Phys. 54 (2013) 063101 (**not obtained**; abstract only).

[5] T. H. Havelock, The stability of motion of rectilinear vortices in ring formation, Phil. Mag. (7) 11 (1931) (not read; cited for context only).